

Electronic Mail

Module 18

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Hello friends I am Professor yogesh Chaba from Department of computer science and engineering Guru jambheshwar University of Science and Technology Hisar harana in this lecture we will study about email system first of all please see the contents which I will cover after covering brief introduction about email system I shall take up email message format which will be followed by message communication protocol SMTP in last message reading protocols POP 3 IMAP and mapi will be explained email which is short form of electronic mail consist of digital messages which are sent and received using the internet it is a method of exchanging digital messages from one sender to one or more recipient email operates across the

internet or other computer networks some early email systems required the author and the recipient to both be online at the same time today's email systems are based on a store and forward model email servers accept Forward deliver and store messages neither the users nor their computers are required to be online simultaneously an email system support five basic functions let us see one by one composition it refers to the process of creating messages and answers although any text editor can be used for the body of the message the system itself can provide assistance with with addressing and the numerous header Fields attached to each message transfer which refers to moving messages from the originator to the

recipient reporting it has to do with telling the originator what happened to the message was it delivered was it rejected or was it lost displaying displaying of incoming messages is required so that people can read their email disposition which is the final step and concerns what the recipient does with the message after receiving it possibilities include throwing it away before reading throwing it away after reading saving it and few others they normally consist of two subsystems the user agents which allow people to compose and read email message format and sending is also taken care by user agent the user agents are local programs that provide a command based menu based 4 minutes, 1 second or graphical method for interacting with the email system protocols which

supports message formatting are RFC 822 and multi-purpose internet mail extensions that is mine which was proposed in RFC 1341 for message reading there are three protocols that are POP 3 IMAP and mapi second subsystem that is message transfer agents is responsible for actual transmission of messages from the source to destination the message transfer agents are typical system demons that is processes that run in the background 5 minutes, 1 second their job is to move email through the network system protocols that support message communication is SMTP that is simple mail transfer protocol first we will see message format protocol for email that is mine the internet email message format defined by RFC

5322 with multimedia content attachments defined in RFC 2045 through RFC 2049 collectively called multi-purpose internet mail extensions or MIM RFC 532 replaced the earlier RFC 282 in 2008 and RFC 282 in 2001 replaced RFC 82 which has been the standard for internet email for nearly 20 years internet email messages consist of two major sections the message header and the message body the header is structured into fields such as from to CC subject date and other information about the email the body contains the message as unstructured text sometimes containing a signature block at the end the header is separated from the body by a blank line now first let us see message header of RFC 82 and mine messages consist of a primitive

Envelope as described in RFC 821 with some number of header Fields a blank line and then the message body each header field logically consist of a single line of text containing the field name a colon and a value for each field RFC 82 was designed decades ago and does not clearly distinguish the envelope fields from the header Fields the principal header Fields related to message transport are listed as shown in figure the two field gives the DNS address of the primary recipient having multiple recipients is also allowed the CC field gives the address of any secondary recipients in terms of delivery there is no distinction between the primary and secondary recipients the BCC that is blind carbon copy field is like the CC field except

that this line is deleted from all the copies sent to the primary and secondary recipients the next two field from and sender tell who wrote and sent the message respectively these need not be the same a line containing received is added by E each message transfer agent along the way this line contains the agent's identity the date and time the message was received and other information that can be used for finding bugs in the routing system the return path field is added by the final message transfer agent to tell how to get back to the sender in addition to the fields discussed just now RFC 82 messages may also contain a variety of header Fields used by the user agents or human recipients the most common ones are listed in figure as shown they are date

which is used to indicate date and time of the message sent reply to indicates email address to which replies should be sent message ID indicates unique number for referencing this message later in reply to indicates message ID of the message to which this is a reply

reference field indicates other relevant message IDs keywords indicates user chosen keywords and subject field indicates short summary of the message in one line display in the early days email consisted exclusive text messages written in English and pressed in aski for this environment RFC 82 explained did the job completely it specified the headers but left the content entirely up to the users nowadays on the worldwide internet

this approach is no longer adequate because sender want to send audio image text video attachment with the mail a solution was proposed in RFC 1341 and updated in RFC 2045 2049 RFC 2045 to 2049 is collectively defined as RFC 532 this solution called mime multi-purpose internet mail extensions is now widely used mime defines five new message headers as shown in figure the first of these is mime version that simply tells the user agent receiving the message that it is dealing with a MIM message and which version of mime it uses the content description field is an asky string telling what is in the message the content ID field identifies the content the content transfer encoding tells how the body is wrapped

for transmission through a network that may object to more characters other than 13 minutes, 1 second letters numbers and punctuation marks the last header shown is really the most interesting one and that is content type it specifies the type and format of the content the type and subtype in this field are separated by a slash for example video video/ MPEG various types and subtypes of content type are shown in figure various types are text image audio video application message and multi-art after message header it comes message body a message body formatted according to the mime 14 minutes, 1 second specification is subdivided into parts that are organized as a hierarchy as with any hierarchy located at the top is the root body part labeled

as R some parts contain other parts known as descendants this figure has two direct descendant body parts label A1 and A2 body part A1 do1 and A1 do2 that do not contain other body parts are defined as content body parts and body part A1 and A2 that do not contain other body parts are defined as multi-art body parts contents body part contain the actual content of the message such as the plain text version of the message and attachment a HTML page and so on so this was about message format of emailing system now we will see how these messages are transferred or communicated the message transfer system is concerned with relaying messages from the originator to the recipient the simplest way to do this is to establish a transport connection from

The Source machine to the destination machine and then just transfer the message simple mail transfer protocol SMTP is an internet standard for electronic mail transmission first defined by RFC a21 in 1982 it was last updated in 2008 with the extended SMTP editions by RFC 5321 which is the protocol used worldwide today SMTP is a connection oriented text based protocol in which a mail sender communicates with a mail receiver by issuing command and supplying necessary data over a reliable ordered data stream channel typically at TCP connection mail processing model using 17 minutes, 1 second SMTP is as shown in figure

email is submitted by a mail client muua that is mail user agent to a mail server

MSA that is mail submission agent using SMTP on TCP Port 587 most mailboxes provide ERS still allow submission on traditional Port 25 from there the MSA delivers the mail to its mail transfer agent MTA often these two agents are just different instances of the same software launched with different options on the same machine the boundary MTA has to locate the target host it uses the domain name system DNS to look up the mail exchanger record MX for the recipient's domain which is part of the email address on the right of at the rate sign the returned MX record contains the name of the target host the MTA next connects to the mail exchanger MX as an SMTP client once the MX accepts the incoming message it hands it to a mail delivery agent

MDA for local mail delivery an MDA is able to save messages in the relevant mailbox format again mail reception can be done using many computers or just one an MDA may deliver messages directly to storage or forward them over a network using SMTP or any other means including the local mail transfer protocol lmtpt a derivative of SMTP designed for this purpose mail is retrieved by end user applications called email clients using IMAP POP 3 or IM AI protocol which we will study in next section this was how SMTP transmission takes place now let us see SMTP session an SMTP session consists of commands originated by an SMTP client that is sender and corresponding responses from the SMTP server that is receiver session is opened and session

parameters are exchanged a session may include zero or more SMTP transactions an SMTP transaction consists of three command reply sequences as shown in figure maale command to establish the return address or return path or bounce address or envelope sender this is the address to which bounce messages should be sent rcpt command to establish a recipient of this message this command can be issued multiple times one for each recipient these addresses are also part of the envelope data consist of a message header and a message body separated by an empty line now we shall discuss protocols for reading of emails interactions between email servers and clients are governed by email protocols the three most common

email protocols for interaction or reading emails by user are post office protocol version 3 POP 3 internet message access protocol IMAP and message application programming interface M AI most email software operates under one of these protocols now let us see each one by one first we shall take up POP 3 the basic POP 3 procedure is to retrieve all inbound messages is for storage on the client delete them on server and then disconnect the user get the email from the ISP's message transfer agent protocol POP 3 described in RFC 1939 was originated with RFC 1081 its current specification is RFC 1939 updated with an extension mechanism RFC 249 and an authentication mechanism in RFC 1734 POP 3 begins when the user starts the mail reader the mail reader

calls up the ISP and establishes a TCP connection with the message transfer agent at Port 110 once the connection has been established the pop three protocol goes through three states in sequence first authorization second transaction third update the authorization State deals with having the user log in the transaction State deals with the user collecting the emails and marking them for deletion from the mail box the update State actually causes the emails to be deleted this Behavior can be observed by typing something like `tnet mail. isp.com Port 110` here `mail. isp.com` represents the DNS name of isp's mail server `tnet` establishes a TCP connection to Port 110 on which the pop three server listens upon accepting the TCP

connection the server sends an asky message announcing that it is present present client starts authorization state by sending over its username and password after a successful login the client can then send over the list command which causes the server to list the contents of the mailbox one message per line giving the length of that message the list is terminated by a period Then the client can retrieve messages using the RR command and mark them for deletion with D E when all messages have been retrieved and possibly marked for deletion the client gives the quit command to terminate the transaction State and enter the update state when the server has deleted all the 26 minutes, 1 second messages it sends a reply and breaks the TCP

connection now after po3 let us see IM map for a user with an email account which is always assessed from one PC pop three works fine and is widely used due to its simply it and robustness but this is always not the case many time user want to assess email from some other computer also which is not possible in pop three this disadvantage gave rise to an alternative delivery protocol IMAP that is internet message assess protocol Define find in RFC 2060 the current version IMAP version 4 is defined by RFC 3501 IMAP is a newer protocol and oriented towards a connection mode of operation the standard IM procedure is to leave messages on the server instead of keeping copies so email is only accessible when

online IMAP does not exclude keeping copies on the client but in an inversion of the way POP 3 works it's the servers copies that are considered the real ones that offers an important security benefit as you won't lose your email if for some reason client computers storage media fails an IMAP server typically listens on well-known Port 143 IMAP provides extensive mechanisms for reading messages or even parts of the messages with large audio and video attachments since the working assumption is that messages will not be transferred to the users's computer for permanent storage IMAP provides mechanisms for creating destroying and manipulating multiple mailboxes on the server in this way a user can maintain a mailbox for each

correspondent and move messages there from the inbox after they have been read after IMAP it is now turn for last email reading protocol that is mapi the message application programming interface M AI is a proprietary email protocol of Microsoft that can be used by Outlook a Microsoft email client software to communicate with Microsoft Exchange which is

its email server software it provides somewhat more functionality than an IMAP protocol unfortunately as a proprietary protocol it works only for Microsoft Outlook Exchange interactions so friends this was about email system used in Internet nowadays in this lecture first we covered introduction of email system which was followed by email message format and message transmission protocol

SMTP in last message reading protocols POP 3 IMAP and mapi were explained thank you